Defects in the published proofs of the classification of homogeneous plane continua, ordered by severity

Working document compiled from the defects ledger of the formalization

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Abstract

The classification of homogeneous plane continua (Hoehn and Oversteegen, 2016) and the papers it rests on have been formalized in Lean 4 with Mathlib. Every statement of the classification, in its plane form and in the form with Bing and Jones's third class, is verified. This document lists what the formalization found wrong or missing in the published texts, most serious first. Two counts are kept: 23 printed statements are false, and 18 printed proofs fail at an identified step while the statement is true. The counts measure kind, not weight. Fifteen of the 23 false statements fail only at an empty set, a single point, an endpoint or a chain of at most three links; in ten of the 18 failed proofs the argument stands once one step is repaired or a small case is set aside. The defects that matter most are in neither count: the step from Jones's decomposition to the circle of pseudo-arcs, used without proof or citation in the classification paper, whose first proof found is in a 2022 preprint of Domokos and Prajs; and Bing and Jones's sketch of a plane circle of pseudo-arcs.

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1 Reader's guide

What is listed. Each item is a defect of a published text, found while formalizing it. Items are numbered consecutively through the document. Each item gives the source and page, the printed text where it matters, what fails, how the formalization settles it, and a status line in italics. The status line names the label from the list below and the section of the project's defects ledger ("Ledger A.23") where the entry is recorded in full.

Labels.

False as printed
The printed statement is refuted by a verified counterexample.

Proof defective; statement true
The printed proof fails at an identified step, and the statement is proved by a repaired or different argument.

Not established
An assertion of a proof sketch that the formalization neither proves in the generality stated nor refutes.

Unstated step
The proof uses a fact the paper neither states nor proves. The fact is true and is proved in the formalization.

Unargued
A sentence of a proof is asserted without argument. It is supplied, or shown unnecessary, in the formalization.

Stated without proof
No proof is printed.

Citation
A cited source does not contain, or does not prove, what it is cited for.

Harmless
Wording that disagrees with itself or with a cited source, without consequence.

Only the first two labels are counted. A defective proof is counted only when its statement is settled; Oversteegen–Tymchatyn's Theorem 3.2 (Item 7) is therefore not counted. A refuted printed sentence is not counted when the same paper states it correctly elsewhere (Hoehn–Oversteegen's restatement of Theorem 1 on p. 211) or when it is a remark and not a numbered statement (their remarks of pp. 193–194); both are listed in §8 and §9. A tacit hypothesis is counted only when a counterexample to the printed form is verified.

Verified. A Lean theorem is *verified* when it compiles and its axiom audit lists only `propext`, `Classical.choice` and `Quot.sound`. Every Lean name in this document is listed in Appendix B

with its namespace and the axioms it depends on, as recorded in the repository's `axioms.txt` (4030 lines). A *hand argument* is an argument written out and read, but not formalized; it is marked as such wherever it occurs.

Quotations are transcribed from page images rendered at 200 dots per inch or more (400 for subscripts, primes and bars), not from the text layer of the PDF files. Page numbers are journal pages. Printed notation is kept: in the older papers $A + B$ is the union and $A \cdot B$ the intersection.

Order. The sections run from the most serious defects to the most trivial:

Section	Kind	False	Proofs
3	Gaps on the path of the classification	0	5
4	Statements false in configurations their authors meant to cover	3	1
5	Other proofs that do not go through	0	2
6	Citations that do not carry what they are cited for	0	0
7	Proof steps that fail locally	0	10
8	Statements false because of their wording	5	0
9	Statements false at degenerate cases	15	0
10	Unargued steps and statements without proof	0	0
11	Misprints and harmless mismatches	0	0
	Total	23	18

Section 3 concerns the published proofs of the results the classification is assembled from: Hoehn and Oversteegen's Theorems A, B and C and their Theorem 2, the arc step that the proof of Theorem B needs, the third class, and Bing's construction of the pseudo-arc. Section 5 concerns supporting results of plane topology used inside those proofs. Moore 1919 Theorem D is of the second kind; it is placed in §4 beside Theorem C, whose false case its proof uses. Within a section, items are ordered by their consequence for the classification. Appendix A lists readings of printed statements that the formalization refuted and that are *not* defects of the sources, among them Bing 1951 Theorem 8 (Item 49).

2 The classification and what it rests on

Hoehn and Oversteegen [HO16, pp. 179–180] assemble the classification from four results:

> Theorem A. ([16]) *If M is a decomposable homogeneous continuum in the plane, then M is a circle of mutually homeomorphic indecomposable homogeneous continua.*
>
> Theorem B. ([12]) *Every indecomposable homogeneous plane continuum is hereditarily indecomposable.*
>
> Theorem C. ([45]) *Every homogeneous indecomposable plane continuum has span zero.*
>
> Theorem 1. *A non-degenerate continuum X is homeomorphic to the pseudo-arc if and only if X is hereditarily indecomposable and has span zero.*

Here [16] is Jones 1955 [J55], [12] is Hagopian 1976 [H76], and [45] is Oversteegen and Tymchatyn 1982 [OT82]. Theorem 1 is the paper's own. Then [HO16, p. 180]:

> It follows immediately from Theorems B and C above and Theorem 1 that every indecomposable non-degenerate homogeneous plane continuum is a pseudo-arc. Combining

this with Theorem A above, we obtain the following classification of homogeneous plane continua.

THEOREM 2. *Up to homeomorphism, the only non-degenerate homogeneous continua in the plane are (1) the circle, (2) the pseudo-arc, and (3) the circle of pseudo-arcs.*

The formal statement. `ho16_classification_plane`$_4$: there are plane sets C_1, C_2, C_3, homeomorphic respectively to the circle, a pseudo-arc and a circle of pseudo-arcs in Bing and Jones's sense (a circle-like continuum with a continuous decomposition into pseudo-arcs whose decomposition space is a simple closed curve), each homogeneous, no two homeomorphic, such that a compact connected plane set with more than one point is homogeneous if and only if it is homeomorphic to one of them. `ho16_classification`$_4$ states the same for a continuum given as a subspace of the plane.

Where each source enters. The formalization reads each cited result in its source and proves it, or proves what the classification needs from it.

- Theorem A: Jones 1955 Theorems 1 and 2 [J55], with Jones 1948, 1949, 1951 [J48, J49, J51], Whyburn 1939 §14 and Whyburn's *Analytic Topology* [W39, W42], Moore 1922 [M22], Moore 1925 [M25] for the plane quotient, and the Effros theorem in van Mill's form [vM04]. Items 5, 11, §7, §9.
- Theorem B: Hagopian 1976 [H76], with Hagopian 1975 [H75]. The formalization proves it by Hagopian 1984 [H84] and the arc step of Bing 1960 [B60], which it proves by a different argument from Jones 1975's [J75]. Items 3, 4.
- Theorem C: Oversteegen and Tymchatyn 1982 Theorem 3.6, with Cook 1967 [C67]. Item 7.
- Theorem 1: Hoehn and Oversteegen 2016 [HO16], with Bing's characterization of the pseudo-arc [B48, B51a, B51b] and its supports [L61, F66, F69, I68]. Theorem 1 is formalized in the direction and generality the classification uses: a hereditarily indecomposable plane continuum of span zero is snake-like (`ho16_theorem1_plane`, whose hypothesis is Hoehn–Oversteegen's Theorem 20, verified as `ho16_theorem20`); with Theorems B and C, a non-degenerate plane continuum is homogeneous and indecomposable if and only if it is a pseudo-arc (`ho16_indec_homogeneous_iff`). Items 6, 9, and Hoehn–Oversteegen's own entries in §§6–11.
- The third class: Bing and Jones 1959 [BJ59] and Domokos and Prajs 2022 [DP22]. Items 1, 2, 8.
- Plane topology used along the way: Moore 1916, 1919, 1920, 1925 [M16, M19, M20, M25] and Mullikin 1922 [Mu22]. Items 10, 12, §§7–9.

What the final proof uses. The repository's dependency census lists, declaration by declaration, what the two final theorems use: 4689 of the library's 11697 declarations, in 446 of its 863 modules. Their closure, 12480 constants of the library, is accepted by the kernel when it is replayed into an environment built from Mathlib alone. Appendix B marks each declaration cited here that the final theorems use. The formal route differs from the published one in places. It does not use Bing 1948 Theorems 3, 4, 5, 7, 8, 9 or 13–15, Bing 1951 (Duke) Theorems 8, 9, 11 and 12, Moore 1922, Jones 1948 Theorems 6 and 10, or Hagopian 1976's own proof of Theorem 1, which it replaces by Hagopian 1984 (Items 3, 6, 9, 38). The items are ordered by their place in the published argument.

3 Gaps on the path of the classification

Each item in this section concerns the published proof of one of the results the classification is assembled from (§1). In each, the printed argument is missing or cannot be completed as written, and the result is proved from another source or by a new argument.

Item 1. The step from Jones's decomposition to the circle of pseudo-arcs (Hoehn–Oversteegen 2016, Theorem 2)

Unstated step in the proof of a main theorem; no proof found before Domokos and Prajs's 2022 preprint. Not counted: no printed statement is false. Ledger A.25, A.26, A.27.

PRINTED. Theorem 2 is obtained by "Combining this with Theorem A above" (§2). The circle of pseudo-arcs is defined on pp. 182–183 as a continuum that "is a circle-like continuum which admits an open map to the circle whose point preimages are pseudo-arcs", and p. 183 adds: "Bing and Jones proved that the circle of pseudo-arcs is homogeneous and that it is unique, in the sense that it is the only continuum (up to homeomorphism) with the above properties."

WHAT IS MISSING. For a decomposable homogeneous plane continuum M that is not a simple closed curve, Theorem A with Theorems B, C and 1 gives an open map of M onto the circle whose point preimages are pseudo-arcs (p. 179: "The conclusion of this theorem implies that there is an indecomposable homogeneous continuum X (possibly a single point) and an open map from M to the circle all of whose point preimages are homeomorphic to X"). Bing and Jones's uniqueness applies to *circle-like* continua: their definition (p. 173) reads "It is a circle-like continuum M such that there is a continuous decomposition of M into pseudo-arcs such that the decomposition space is a simple closed curve." That M is circle-like is used and is neither stated nor cited.

WHY IT NEEDS AN ARGUMENT. An open map onto the circle with pseudo-arc fibres does not give circle-likeness. The product of the pseudo-arc with the circle has such a map and is two-dimensional, so not circle-like; Hoehn and Oversteegen mention it (p. 183: "homogeneous but not embeddable in the plane"). Planarity and dimension one do not suffice either. Prajs [P00, Theorem 8, p. 1753]: "There exist a plane s-arc of pseudo-arcs and a plane s-circle of pseudo-arcs", curves with continuous decompositions into pseudo-arcs over an arc and over a simple closed curve; on the same page: "the examples constructed here are triodic, and thus topologically different. Evidently, their elements of the decompositions are not terminal." Circle-like continua are atriodic, so Prajs's plane s-circle is not circle-like. A proof must therefore use homogeneity. What homogeneity supplies is that Jones's elements are terminal: Jones 1955 Theorem 1(c), p. 736: "if g is an element of G and K is a subcontinuum of M which contains both a point of g and a point of $M - g$, then g is a subset of K" (verified, `jones55_c`).

THE LITERATURE BEFORE 2022. Lewis [Lw83, p. 80] asks, as "Question [28]", whether two (homogeneous) one-dimensional continua with continuous decompositions into pseudo-arcs over homeomorphic quotients must be homeomorphic, and adds: "This question has an affirmative answer if all of the decomposition elements in each of G_1 and G_2 are terminal continua." His [28] is the paper published as Lewis 1985, then submitted; p. 80 gives no proof. Lewis 1985 [Lw85, p. 93], the paper that constructs such continua with terminal elements: "It will follow from our construction that all decomposition elements are terminal continua. We still do not know in general whether $\hat{M}$ is unique." Lewis 1996 [Lw96, p. 173] combines an observation of his own with Lewis 1985 (cited there as Theorem 2 [24], an existence and lifting theorem) and concludes: "For each continuum in one class there is a unique corresponding continuum in the other class, obtained either by shrinking out maximal terminal pseudo-arcs or by enlarging each point to a terminal

pseudo-arc." Uniqueness in the enlarging direction is the question Lewis 1985 leaves open. It is not proved in any of the three papers.

THE FIRST PROOF FOUND. Domokos and Prajs [DP22, Theorem 6.1]: "A continuum X is a circle of pseudo-arcs if and only if X admits an open atomic map $f\colon X \to S$ onto a simple closed curve S such that all point-inverses, $f^{-1}(y)$, are pseudo-arcs." Atomic means that every point-inverse is terminal; "circle of pseudo-arcs" means homeomorphic to Bing and Jones's example. The authors: "We include the proof because we did not find in the literature a proper reference to this theorem." The paper is an arXiv preprint; no journal version and no earlier proof were found in a search of the papers on homogeneous continua available to the project. Their proof shows that X is atriodic ("one can easily show"; unargued, and not needed by the formal route below), that the preimages of the two halves of S are atriodic folders of chainable continua, chainable by [HMP20, Proposition 6], and glues them to Bing and Jones's example by Bing–Jones Theorem 10.

FORMALIZATION. Verified, with no named hypothesis: Theorem 6.1 in both directions (`dp22_theorem61'`, `dp22_theorem61_iff'`), the circle-likeness step (`hoCircleLike'`), and the classification (`ho16_classification`$_4$, `ho16_classification_plane`$_4$). The formal route makes the halves chainable by Bing's argument for Theorem 11 of [B51a] (`isSnakeLike_of_atomic'`), which needs neither folders nor shape theory, and glues them by Bing–Jones Theorem 10 (`bjThm10`). Jones's decomposition of M supplies the atomic map through `jones55_c`. In the formalization "circle of pseudo-arcs" has Bing and Jones's meaning (the definition `IsCircleOfPA`: circle-like, with an open map onto the circle whose fibres are pseudo-arcs). Domokos and Prajs's meaning, homeomorphic to Bing and Jones's example, is not formalized, since that example is not constructed; by the two theorems above, any two circles of pseudo-arcs in Bing and Jones's sense are homeomorphic.

Item 2. A plane circle of pseudo-arcs (Bing–Jones 1959, §7, pp. 189–191)

Not established: a sketch whose key step is not argued. Not counted: no instance shows the step failing. The claim is verified by a different construction. Ledger A.25 (sixth reading).

PRINTED. §7 begins (p. 189): "It follows from Theorem 10 that each circle of pseudo-arcs is homogeneous and that any two of them are homeomorphic. In this section we show that the plane contains a circle of pseudo-arcs." On p. 190: "Now there exists a sequence $D_1, D_2, D_3, \cdots$ of circular chains of simple domains in the plane such that" conditions (1)–(9) hold. On p. 191: "With the addition of (9) one may see with the help of [3] that each element of G is a pseudo-arc. It follows from Theorem 10 that M is homogeneous."

WHAT IS MISSING.

- The chains D_i are asserted to exist and are not constructed.
- That the elements of G are disjoint, fill M, and form a continuous decomposition whose quotient is a circle is not argued, with or without (9). Unargued also: "Also, in this case M would be a continuous circle-like circle of continua" (p. 191, of the chains without (9)).
- Not established: that each element of G is a pseudo-arc. Condition (9) makes each new column T_{ij} crooked in a pair $T_{hk} + T_{hl}$ of old columns. It does not control an element of G covered at some level by two adjacent columns, and two crooked chains joined end to end are not crooked in general. No chains satisfying (1)–(9) with an element of G that is not a pseudo-arc are known, so the step is not shown to fail.

CONSEQUENCE. The third class of the classification is a *plane* continuum only through §7. Hoehn and Oversteegen (p. 179): "Bing and Jones [5] constructed in 1954 such a continuum in the plane

for which X is the pseudo-arc, and also proved that it is homogeneous." Lewis [Lw85, p. 91] also takes §7 at face value. His own construction of a circle of pseudo-arcs [Lw85, p. 93] identifies "the end pseudo-arcs of arcs of pseudo-arcs in the same manner that the endpoints of arcs are identified"; it takes the arc of pseudo-arcs from Bing and Jones and does not discuss planarity or circle-likeness.

FORMALIZATION. Verified: `bjPlaneExample`, through `exists_circDescr_plane` and `CircDescr.isCircleOfPA`. The construction is not Bing and Jones's. It uses nested circular chains of plane domains whose links are neighbourhoods of arcs of a base circle; the pattern of each chain in the previous one is the lift of a degree-one map built from *sweeps*, along each of which the window of links changes at one end at a time (`sweep_lemma`, `sweep_lemma3`); and a planar step (`circ_step`) obtained by descending an equivariant shear of a strip through the exponential map. A recognition theorem shows that the intersection is a circle-like continuum with a continuous decomposition into pseudo-arcs over the circle (`CircDescr.isCircleLike_M`). Nothing of §7 is used except its claim.

Item 3. Hagopian 1976, Theorem 1 (Theorem B), pp. 345–349

Proof defective; statement true. Counted (one defective proof). The printed route is not completed; the theorem is verified by Hagopian's 1984 route. Ledger A.23, A.24.

STATEMENT. Every indecomposable homogeneous plane continuum is hereditarily indecomposable. The proof argues by contradiction from a type A′ subcontinuum E, through an induction over the blocks $x_1 < x_2 < \cdots < x_{2n}$ of a continuum F in a strip $\pi^{-1}[D]$, "Following Bing [2, p. 225]" (p. 348), that is, Bing 1960 [B60, pp. 225–227].

THE FAILING STEP. p. 348: "The proof now breaks down into two cases". The case split presupposes that $d^{-1}(x_{2i-1})$ and $d^{-1}(x_{2i-2})$ each meet a side segment $\pi^{-1}[\langle p_1, r\rangle]$ or $\pi^{-1}[\langle p_5, s\rangle]$. At the last step of the induction ($i = 2$, $J = [x_1, x_2] = [0, e]$) this is false: $d^{-1}(x_2) = d^{-1}(e) = k^{-1}(1)$ lies over the far end of $\pi[E]$ and meets neither side segment. So neither printed case applies to the step that produces the final contradiction with "$[x_1, x_2]$ does not have Property 4". Certificate: `blockLayersMeetSides_false`. In Bing's model [B60, p. 226] the corresponding points are end points of arcs irreducible between the two side lines, so they lie on the sides by construction; Hagopian's block ends are values of d, and nothing printed places their layers on the sides. For $i \geq 3$ the presupposition is true (`blockLayersMeetSides3`), and the last step can be read separately with the bottom edge in place of the strand (`last_step_ending`).

A LEMMA FROM THE 1984 PAPER. "Note that $x_1 = 0$, $x_2 = e$, and $x_{2n} = 1$" (pp. 347–348), in its last clause, and Property 4 for the top block need a saturation property the text does not state: every component of $F \cap \pi^{-1}[D]$ through a point over the inner middle of D crosses D. The formalization proves it (`innerComponentsCross_hagopian`), and at the end layer $d^{-1}(1)$ of F only with Hagopian 1984 Lemma 7 (`hagopian84_lemma7`): when two type λ continua overlap without either containing the other, every layer of one that meets the other is a layer of the other. The 1976 paper has nothing of this kind at end layers; its Lemma 3 covers interior layers.

PRINTED SENTENCES NEITHER PROVED NOR REFUTED. By a verified theorem (`hagopian76_theorem1_of_cases4`) the 1976 route reduces to four named claims, which are open. They collect the following sentences of pp. 348–349.

- Case 1: "By considering the position of $d^{-1}[[x_{2i-3}, x_{2i}]]$, we see that y_{i-1} is between $d^{-1}[I]$ and $d^{-1}[J]$". The position argument needs the unicoherence of the plane, which is neither stated nor cited; Hagopian's strands, components of $F \cap \pi^{-1}[D]$, need not end on the side lines as Bing's arcs do, and L may re-enter $\pi^{-1}[D]$ on the far side of a strand.
- Case 1: the segment Q of the vertical over p_4 closes the band between two blocks only if the

band crosses that vertical once.

- p. 348: "the continuum N does not come within ε_2 of $\pi^{-1}(p_4)$". Points of N outside $\pi^{-1}[D]$ are not addressed. Bing's corresponding sentence [B60, p. 227] is unargued as well.
- Case 2 (p. 349): "If y_{i-1} is above $d^{-1}[J]$, it is between $d^{-1}[I]$ and $d^{-1}[J]$, and $f(y_{i-1})$ is above $d^{-1}[I]$" and "Hence y_{i-1} is below $d^{-1}[J]$ and $f(y_{i-1})$ is above". No reason is given. Bing's corresponding sentence follows from the parity of crossings of a simple closed curve (a hand argument on polygons); for Hagopian's continua there is no parity argument.
- p. 348: "(Other cases are handled with similar arguments to that given in this case.)" Of the four orientations, two are treated by the formalization; in the other two the frame is not symmetric (the bottom edge lies over E, the top over I_2).
- p. 348, E2.4: "The point y_{i-1} is below C_i and $\rho(y_{i-1}, d^{-1}[I]) \geq \varepsilon_3$; for otherwise, a subcontinuum of L would violate Property 2" is verified for the strand next to C_i only, conditionally on three sub-claims; the printed form, for all of $d^{-1}[I]$, is not decided.
- p. 349, T17: "Hence $f(y_{i-1})$ is either above $d^{-1}[I]$ or below $d^{-1}[J]$" presupposes that every point off the crossing continuum is above or below it. Components of $F \cap \pi^{-1}[D]$ that meet the frontier only in a side segment are not addressed.

FORMALIZATION. Verified: `hagopian76_theorem1`, by Hagopian 1984 Corollary 3, which carries out F. B. Jones's plan: Hagopian 1984 Theorem 1 with $N = M$ (`hagopian84_decomp`) gives a monotone continuous decomposition with a homogeneous plane quotient containing an arc, and the arc step (Item 4) gives the contradiction. About twenty further unargued steps of pp. 340–349 are supplied (§10).

Item 4. The arc step: Jones 1975, property (f) and the Theorem, pp. 59–61

Two proofs defective; statements true, that of (f) only vacuously in the plane. Counted (two defective proofs). Verified by a different argument. Ledger A.28.

ROLE. Bing 1960 [B60] proves that a homogeneous bounded plane continuum containing an arc is a simple closed curve. Hagopian's 1984 route to Theorem B needs it for indecomposable continua. Jones 1975 [J75] gives a short proof: "THEOREM. *If M is a homogeneous plane continuum containing an arc, then M is a simple closed curve.*" (p. 59).

PROPERTY (F), P. 59. "If p is a point of M and D is an open set in E with $p \notin \bar{D}$, then there exists a positive number ϵ such that no arc abc in M has the property that both a and c belong to D and the ϵ-neighbourhood of the subarc ab contains both p and the subarc bc." The proof ends: "Since $b'c'$ is in the ϵ-neighbourhood of $a'b'$, the image of $b'c'$ must be in the 3ϵ-neighbourhood of the image of $a'b'$. This is impossible since the 3ϵ-neighbourhood of q_1p_1 contains no point of p_2q_2 (and conversely)." This does not follow: the image of $a'b'$ may run along the arc q_1pq_2 past q_1 and come back near p_2q_2. The proof uses only the Effros theorem and property (c), hence only hypotheses that the dyadic solenoid satisfies (homogeneous, indecomposable, hereditarily unicoherent, atriodic, with (c)); in the solenoid (f) fails, since a long arc of a dense line is ϵ-dense (a hand argument, not formalized). Two earlier steps are unargued as well: "takes b to p" and "b' is close enough to b" need b near p, and "leaves the images q_1 and q_2 of a and c respectively in D" is unjustified for a homeomorphism that is not small. For plane continua (f) holds only vacuously, once the Theorem is known (no Lean declaration states (f)).

THE THEOREM, PP. 59–61. The step "This makes h take $[t_0, t_d]$ onto (approximately) $[t_d, t_0]$. So there is a fixed point t_f" is false for small ϵ: a small homeomorphism preserves the orientation

of the ray T (`JonesSetting.orientation`, verified). Under the hypotheses of the setting the step contradicts this orientation lemma, so any argument for it would itself finish the proof. The Theorem also uses (f) twice, and four further steps are unargued (§10).

FORMALIZATION. Verified: `jones75_noArc` (no indecomposable homogeneous plane continuum contains an arc) and `bing60_plane_uncond` (Jones's Theorem). The proof is a Poincaré–Bendixson argument in a collar box around an arc of a line, not Jones's: by Effros and the orientation lemma, points of the line near the base arc lie on left-to-right passes of the box; the region between the base pass and a later thin pass is bounded by a Jordan curve made of a vertical segment and an arc of the line; once the line leaves that region it never re-enters it (`trap_jordan`), which contradicts recurrence (`jones75_contra`). Jones's (d) is verified in both directions (`JonesSetting.jones75_d`), and (c) through `JonesSetting.union_arc` and `JonesSetting.nested`.

Item 5. Jones 1955, Theorem 1 (Theorem A), the case argument of pp. 737–738

Proof defective; statement true. Counted (one defective proof). Ledger A.11.

STATEMENT. p. 736: "THEOREM 1. *Suppose that M is a decomposable, homogeneous, compact, metric continuum. Then there exists a nondegenerate continuous collection G of mutually exclusive continua filling up M such that (a) with respect to its elements (as points) G is a homogeneous, aposyndetic, compact, metric continuum, (b) if x is a point of M, then L_x is an element of G, and (c) if g is an element of G and K is a subcontinuum of M which contains both a point of g and a point of $M - g$, then g is a subset of K.*" Here L_x is x together with the points y at which M is not aposyndetic with respect to x, and $U_x = M - L_x$.

WHAT FAILS. In Case I, "Since M_0 is a subset of M_1" fails as written: when $o(q) \in V_2$, the irreducible continuum $K_o(q)$ can be $\{o\}$, which is not in M_1; the text does not keep $o(q)$ out of V_2. And "Since q does not cut o from U_q" draws on $q \notin N_o$, which is stated with U_o; it therefore needs $U_q = U_o$, which is (b), not yet available at that point. The argument serves $N_o = M - U_o$.

FORMALIZATION. Verified: (a), (b), (c), upper semicontinuity, continuity, and that the elements are homogeneous and mutually homeomorphic (`jones55_a`, `jones55_b`, `jones55_c`, `jones55_usc`, `jones55_continuous`, `jones55_elements`). Jones's claims (1)–(10) and his properties (a)–(f) of pp. 738–739 are kept. The case argument and the upper-semicontinuity argument are replaced by a spreading lemma from the Effros theorem: in a homogeneous compact metric space, a continuum with interior that contains o and misses x gives aposyndesis at o with respect to x (`aposAt_of_isContinuum_interior`).

Item 6. Bing 1948, Theorem 3, the special case, p. 732

Proof defective; statement true. Counted (one defective proof). Ledger A.1.1.

STATEMENT. "THEOREM 3. *If the chain D is crooked in the chain $E = [e_1, e_2, \cdots, e_m]$ and d_i is a particular link of D, then there is a chain F such that F is a consolidation of D, d_i is contained in only the first link of F, each link of F is a subset of the sum of two adjacent links of E, and any link of F containing an end link of D which intersects e_1 or e_m is a subset of $e_1 + e_2$ or $e_{m-1} + e_m$.*"

WHAT FAILS. In the special case (no interior link of D meets e_1 or e_r), with d_i a link of $D(1, h)$, Bing takes H from the induction hypothesis and sets F to be $H(1, u-1)$, followed by the level groups of the remaining links D', followed by $D(k+1, n)$. Certificate (`Recon121.seam_every_H_fails`, with `Recon121.seamD_special`): eleven links of a realized chain over five coarse links satisfy the special-case hypotheses, with $(h, k) = (4, 6)$ forced; a chain H satisfying the induction hypothesis exists; yet for every $i \leq h$ and every such H, the chain Bing writes down is not a chain with d_i in its first

link only.

FORMALIZATION. Verified: the theorem as printed in every clause (`theorem3_literal`, `Recon121.theorem3LiteralOr_holds`, `theorem3_bing`). The repair (`Recon122.bing_main`) makes three changes, none in the text: the level groups are anchored at the bottom; an explicit grouping is made when there are five coarse links (`Recon122.special_five`); and the branch $d_i \in D(h,k)$ is handled by the induction hypothesis applied to $D(h,k)$ inside $E(2, r-1)$.

Item 7. Oversteegen–Tymchatyn 1982, Theorem 3.2, pp. 137–138, in the proof of Theorem C

Proof defective; statement not decided. Not counted. Ledger A.29.

PRINTED. The proof begins: "We may suppose that for each $\lambda \in \Lambda$ $h_\lambda(Y)$ meets $\{0\} \times R$ in a single point and $h_\lambda(Y)$ intersects $\{1\} \times R$ in at most one point. Hence, … we may suppose that for each $\lambda \in \Lambda$ neither $h_\lambda(X)$ nor $h_\lambda(Y)$ separates the plane. Let $\delta \in \Lambda$. We may suppose that $\pi_1(h_\delta(y)) > \pi_1(h_\delta(z)) + 1/2$ for each $z \in X$. We may also suppose that $d(h_\lambda(z), h_\gamma(z)) < 1/8$ for each $z \in Y$ and $\lambda, \gamma \in \Lambda$. The proof now proceeds as in Theorem 3.1".

WHAT FAILS. The two suppositions about δ are not available together: the cluster of maps within $1/8$ need not contain δ, and one rescaling does not fix the other gaps. The single-point supposition is unargued and can be avoided but not justified (disjoint crossing arcs can each meet a side in two points). The proof of Theorem 3.1 picks three images one of which separates the other two in the strip, which is unargued.

FORMALIZATION. What Theorem C needs is a strip form in which the images lie in disjoint continua crossing the strip, with their traces on the side ordered. That form is verified (`ot82_theorem32`). From it, Corollary 3.5 and Theorem 3.6 in the indecomposable case follow (`ot82_cor35_of_thm32`, `ot82_theorem36_of_cor35`, two implications), and Theorem C is verified with no hypothesis (`theoremC`). Whether the printed Theorem 3.2 holds in general is not decided.

4 Statements false in configurations their authors meant to cover

Item 8. Bing–Jones 1959, Theorem 5, pp. 175–176

False as printed. Counted (one false statement). Its one use is existential and survives. Ledger A.25 (second reading).

PRINTED. "Then if U is an open set such that $U \cdot G \neq 0$, $U \cdot (g_0 + g_1) = 0$, and $\bar{U}$ lies in a link of E, then $G - U$ may be expressed as the sum of two mutually separated sets A, B such that $g_0 \subset A$, $g_1 \subset B$, and for each x $(0 \leq x \leq 1)$, either $A \cdot g_x$ intersects each link of E or $B \cdot g_x$ intersects each link of E."

WHAT FAILS. In every instance with $n \geq 3$ links. Merge the last two links (replace e_{n-1} by $e_{n-1} \cup e_n$ and keep e_n), so that $\overline{e_n} \subseteq e_{n-1}$; the merged collection covers G, properly covers every fibre, and is a chain in Bing and Jones's sense, with nonadjacent links at positive distance (checked by hand; the certificate uses the library's notion of chain, in which nonadjacent links are disjoint). Take U a small neighbourhood of $\overline{g_c \cap e_n}$ inside e_{n-1}. Then $g_c - U$ misses e_n, and the conclusion fails at c for every separation, although no fibre lies in U. Certificates: `bj_theorem5_printed_false_general` (from every instance with three or more links) and `bj_theorem5_printed_false` (a fibre inside U). The printed proof passes from separations of $G - d_j$ to separations of $G - U$.

CONSEQUENCE. The theorem's one application (p. 185, in Step 2(c) of Theorem 10: "By Theorem 5 there is an open set U_i …") is existential, and a small enough U_i exists. Theorem 10 is unaffected. The repaired form asks that f be open and that no fibre meet a link only inside $\bar{U}$; it is verified (`bj_theorem5`, `theorem5_repaired`).

Item 9. Bing 1948, Theorem 8, p. 737

False as printed. Counted (one false statement). Theorem 9's proof uses it. Ledger A.1.3.

PRINTED. "THEOREM 8. *Suppose the chain D_n $(n = 1, 2)$ is crooked in the chain E, each end link of E contains a link of D_n, and no proper consolidation of D_n has these properties. Then D_1 follows a pattern in E that D_2 follows in E, and the chain whose links are the sums of corresponding links of D_1 and D_2 is crooked in E.*" The proof: "That the chain whose links are the sum of the corresponding links D_1 and D_2 is crooked in E may be proved by induction on the number of links in E."

WHAT FAILS. The unions of corresponding links need not form a chain. Certificate (`not_theorem8_literal`): in a finite discrete subspace of $\mathbb{N}$, two chains of six links (the fewest possible, `theorem7_minimal`) are crooked in a four-link chain E, have their first links in e_1 and last links in e_4, and follow Bing's pattern in E in the same direction; the second and fourth unions meet (at the point 20), so the unions form no chain, and pairing the links in opposite directions gives unions that do not refine E. Bing's chains have open links that need not be connected (p. 729: "We have not supposed that the links of a chain are necessarily connected"), so the discrete example applies; minimality holds because a proper consolidation would have fewer links. The mechanism is that the pattern revisits levels, so links of D_1 and D_2 at non-adjacent positions can meet.

CONSEQUENCE. The proof of Theorem 9 (p. 737) begins "By Theorems 7 and 8 there are chains G_1 and G_2 … and the chain H … whose links are the sums of corresponding links of G_1 and G_2 is crooked in E." The corrected Theorem 8 (`Fold.theorem8`, with the orientation made explicit and the chain property of the unions as a hypothesis) suffices. Theorem 9 is verified with its mesh clause weakened: each link of F lies in the sum of three consecutive links of E, where Bing has "the sum of two adjacent links of E" (`Fold.theorem9`). The printed two-link form is not formalized.

Item 10. Moore 1919, Theorems C and D

Theorem C false as printed; the proof of Theorem D defective, statement true. Counted (one false statement, one defective proof). Ledger A.15.

THEOREM C. It concerns rulings of a closed curve with respect to a quadrilateral $ABCDA$. A closed curve that touches a side from within meets $ABCDA$ in one point and admits no ruling of the required kind. Moore offers an "Indication of proof … given instead of a tedious formal proof covering all possible cases", which misses this case. Certificate: `not_m19_theoremC_printed`, in every model of Moore's Axioms 1–8. Three corrected forms are verified: absorbing finitely many disjoint cross-cuts (`m19_theoremC_crosscuts`), a curve within $ABCDA$ (`m19_theoremC_within`), and the printed statement with touching from within excluded (`m19_theoremC_corrected`). Touching from without is harmless.

THEOREM D. Moore's proof applies Theorem C to a curve $\bar{J}$ that g_1 may touch, so it rests on the false case of C; its induction over $g_2, \ldots$ needs D itself; and the construction of Fig. 2 is unargued ("It easily follows"). Verified (`m19_theoremD`) by an induction over a finite subcover: at each step a curve is built about the uncovered part, crossing $ABCDA$ at every common point, and the corrected Theorem C is applied. Whether Moore's $\bar{J}$ could be re-chosen so that it is crossed and not touched

is not decided.

Consequence. The paper's conclusion, that a space satisfying Moore's Axioms 1–8 is homeomorphic to the plane, is verified (`m19_plane`), with Theorems E, G and the final paragraph reached by a route different from Moore's (§10).

5 Other proofs that do not go through

The items of this section concern supporting results of plane topology used inside the proofs of §3. In each, the printed argument fails in a case the statement covers, and the result is proved by a different argument.

Item 11. Whyburn 1939, Theorem 14, p. 747

Proof defective; statement true. Counted (one defective proof). Ledger A.21.

Printed. "Clearly we may suppose R bounded, since an inversion of the plane can be used to throw R into a bounded region complementary to the image of M."

What fails. The centre of the inversion must lie in a complementary domain of M other than R. When M has no bounded complementary domain, for instance when M is a closed disk or any continuum that does not separate the plane, there is none. This is a natural case of the theorem, not a degenerate one.

Formalization. Verified for bounded and unbounded domains alike, without inversion (`whyburn39_theorem14`). Moore's curve C is replaced by a joining lemma from Janiszewski's theorem, and Whyburn's curve K and limit continuum H by two facts: a point where infinitely many disjoint connected sets accumulate has no finite order, and Whyburn's (3.2).

Item 12. Moore 1925, Theorem 21: Axioms 6 and 7

Proof defective; statement true. Counted (one defective proof). Ledger A.16.

Printed. "With the aid of results established in my paper *Concerning a set of postulates for plane analysis situs* … it easily follows that Axiom 6 and 7 also hold true."

What fails. Moore 1919 proves nothing for the axiom system Σ_2 beyond the assertion of its introduction. The step therefore rests on the equivalence of Σ_1 and Σ_2 that Moore 1916 asserts on p. 163 without proof ("It is easy however to show that though Σ_1 and Σ_2 are not absolutely equivalent, they are equivalent with respect to point and limit point"). The formalization neither proves nor needs that equivalence.

Formalization. Verified (`m25_theorem21`): Axioms 6 and 7 are proved directly, with cross-cuts and Moore 1925 Theorems 10, 12, 15, 18 and 19. The other three "easily" claims of the proof are confirmed. Moore 1925 Theorem 22, the plane quotient used for Theorem A, is verified from Moore 1919 for Σ_1 only (`m25_theorem22`).

6 Citations that do not carry what they are cited for

None of these is counted. The first bears on the classification (Item 1); the others were avoided or repaired.

Item 13. Lewis 1983 and 1996: uniqueness attributed to work that leaves it open

Citation: the cited work leaves the claim open. Ledger A.27.

Lewis [Lw83, p. 80] poses the uniqueness question as "Question [28]", where [28] is Lewis 1985 (then submitted), and asserts without proof that it "has an affirmative answer if all of the decomposition elements in each of G_1 and G_2 are terminal continua". Lewis [Lw96, p. 173] asserts the uniqueness of the continuum "obtained either by shrinking out maximal terminal pseudo-arcs or by enlarging each point to a terminal pseudo-arc", on the strength of an observation of his own and of Lewis 1985, Theorem 2 [24], which gives existence and lifting only. Lewis 1985 [Lw85, p. 93]: "We still do not know in general whether $\hat{M}$ is unique"; and [Lw85, p. 99, Question 2]: "If $\hat{M}_1$ and $\hat{M}_2$ are two continuous curves of pseudo-arcs with homeomorphic decomposition spaces, are $\hat{M}_1$ and $\hat{M}_2$ homeomorphic?" The case the classification needs is Domokos–Prajs Theorem 6.1 (Item 1).

Item 14. Domokos–Prajs 2022, Example 6.1: terminality of Bing and Jones's decomposition

Citation: the cited work does not prove the claim; not used. Ledger A.27.

p. 13: Bing and Jones "constructed the *circle of pseudo-arcs*, denoted here by Ψ, which is a homogenous, decomposable and nowhere locally-connected continuum in the plane admitting a continuous decomposition into terminal pseudo-arcs." Bing and Jones do not prove that the elements are terminal. This supports "It is known" in the proof of the "only if" direction of Theorem 6.1 (p. 14). The formalization proves that direction differently: a circle of pseudo-arcs is homogeneous (Bing–Jones Theorem 10), so Jones's decomposition gives an atomic map (`jones55_c`), and the property passes to every homeomorphic space (`dp22_theorem61_iff'`).

Item 15. Hoehn–Oversteegen 2016, pp. 191 and 211: a research announcement cited for a proof

Citation: an announcement without proofs; avoided. Ledger A.26.

p. 191: "Because X has span zero, it is tree-like [31]"; p. 211: "Since X has span zero, by [31] we have that X is tree-like." Here [31] is Lelek 1979 [Le79], a research announcement (p. 631: "The purpose of the present paper is to announce some results … A complete version will be published elsewhere"). Its Propositions 1–4 are called "rather straightforward" and are not proved; its Lemma 5 and Theorem 6 carry no proof; its Corollary 9 drops Theorem 6's hypothesis "$\dim X \leq 1$"; its Corollary 10, that span zero implies tree-like, is obtained in one paragraph through three citations. On p. 212 Hoehn and Oversteegen cite a published proof of a stronger statement ([46], Oversteegen–Tymchatyn 1984). The formalization does not need tree-likeness: for plane continua a grid graph serves, since hereditary indecomposability gives empty interior and radial projection in the squares of a fine grid maps the continuum into the union of their edges.

Item 16. Fugate 1966: Miller's and Sorgenfrey's theorems cited in over-general form

Citation: over-general as cited; harmless in context. Ledger A.4.3, A.4.4.

As cited, Miller [6, Theorem 1.5, p. 180] says: if $A \cup B$ contains a non-unicoherent subcontinuum D, then $D \subseteq A$ or $D \subseteq B$. This is false: the closed disk is the union of two half-disks, and its boundary circle lies in neither (`miller_1_5_false`). What Fugate needs is hereditary unicoherence of $A \cup B$ for chainable A, B with connected intersection, proved directly (`isHereditarilyUnicoherent_union`). Miller's own statement has not been checked against the citation. As cited, Sorgenfrey [8, p. 440]

says that three intersecting continua, none contained in the union of the other two, have a union that is a triod; a circle is the union of three arcs. Inside a hereditarily unicoherent continuum, which is Fugate's situation, it holds. In Fugate's setting the continua are also a-triodic, and neither citation is needed in the generality printed.

Item 17. Hagopian 1975, 1976 and 1984: citations that do not cover the step

Citation: the cited result does not cover the step; each gap filled. Ledger A.22–A.24.

- Hagopian 1976, p. 344: "Since M is indecomposable and homogeneous, no proper subcontinuum of M separates R^2 [5, Lemma 1 (second paragraph of proof)]". The cited argument carries a *non-unicoherent* continuum to every composant; for a separating one it needs Borsuk's theorem that plane separation is a topological invariant, which neither paper states. Verified without Borsuk (`no_proper_subcontinuum_separates`, through `exists_separating_subcontinuum_of_homeomorph`).
- Hagopian 1984, p. 346: "since the elements of $\mathfrak{D}$ are tree-like, if M is planar, then $M/\mathfrak{D}$ is planar [21]". Moore 1925 needs elements that do not separate the plane; that tree-like continua do not is not stated. Filled in the plane by the same theorem.
- Hagopian 1975, Lemma 2 (pp. 36–37): "[10, Theorem 3, p. 216]" states that Kuratowski's family has the order type of the interval, not that the continuum is of type A′. Filled by a countable dense subfamily, Cantor's theorem and the map $k(y) = \inf\{q : y \in D_q\}$. Hagopian 1984 p. 335 repeats the citation with the same gap. Both also cite Thomas 1966 (Theorem 10, p. 15), which was not in hand and is not checked; the gap is recorded for the Kuratowski citation only.
- Hagopian 1975, paragraph 13: "[9, Theorem 2]": Jones 1955 Theorem 2 implies, but does not state, that a unicoherent homogeneous plane continuum is indecomposable. Supplied.
- Hagopian 1984, Lemma 9: the citation (1975, paragraphs 5–12) treats interior layers; the lemma claims every element, end sets included. True through an extension, not stated.

Item 18. Other citations

Citation; harmless. Ledger A.11, A.18, A.21, A.24.

- Jones 1955, Theorem 2, last step: "Hence by [5]": the result used is [3] (Jones 1949), not [5] (Jones 1951).
- Jones 1949, reference [8]: "Theorem 9.1, p. 61" is Whyburn's (9.2); (9.1) is the lemma before it. Reference [4]: Menger's theorem characterizes the circle by *all* points being of order 2, and Jones applies it to *a* point of order 2; homogeneity and the invariance of order are tacit (supplied, `jones49_order_two`).
- Whyburn 1939, p. 748, footnote 14: "by a theorem of R. L. Moore" gives no theorem number. In substance it is Moore 1916 Theorem 43 in corrected form (Item 34).
- Hagopian 1984, p. 346: "In 1968 F. B. Jones [16]", where the reference list gives 1969.

7 Proof steps that fail locally

In each counted item of this section a step of the printed proof is false as written, but the argument stands for the main case: a short argument inside the same proof repairs the step, or (Item 19) the failure is confined to chains of at most three links, which need another construction.

Item 19. Bing–Jones 1959, Theorem 7, proof, p. 177

Proof defective; statement true. Counted (one defective proof). Ledger A.25 (third reading).

The reduction to the chain $D' = U_0, U_1, U_2, d_3, \cdots, d_{n-2}, U_{n-1}, U_n, U_{n+1}$ works for $n \geq 4$. For $n = 3$ its conditions cannot all hold: a link serving as both U_2 and U_{n-1} contains $P \cdot d_2$ and meets U_4 (certificate `bj_theorem7_proof_n3`). Chains of one or two links are not treated. Unargued as well: "We suppose with no loss of generality that $r < s$"; the special case $A \subset e_1$; the existence of the crooked sequence $D_1 = D, D_2, \cdots$; and the non-emptiness of $P \cdot (d_1 - d_2)$. Verified by another construction, following the pattern along a walk and folding, with no special case (`bj_theorem7`); the second sentence holds for $n \geq 3$ with a collar chain of $n + 4$ links in place of D'.

Item 20. Hoehn–Oversteegen 2016, Proposition 19, p. 208

Proof defective; statement true. Counted (one defective proof). Ledger A.26 (fourth reading).

"If $D \subset F_1 \cup F_2$, then … $\varphi(D)$ is contained in a component $\widetilde{D}$ of $G \setminus \pi(S_{i_0})$. Moreover, $\varphi(D) \subset \pi(P_1 \cup P_2)$, so $\partial\widetilde{D} \cap \pi(\alpha_{i_0}) = \emptyset$ by (S6′) for S. This means that either $\partial\widetilde{D} \subset \pi(\beta_{i_0})$ or $\partial\widetilde{D} \subset \pi(\gamma_{i_0})$." The middle sentence is false. Certificate `ho16_p208_false`, verified on the graph G: a broken stairwell structure of height 3 with its pit at level 2 on $G = [0, 10]$, every sheet at height $\frac{1}{2}$, with $\pi(S_1) = [4, 10]$, $\pi(S_2) = [4, 6]$, $\pi(S_3) = G$, $\pi(\alpha_2) = \{4\}$, $\pi(\gamma_2) = \{6\}$, $\pi(P_1) = [1, 2] \cup [\frac{11}{2}, 10]$, $\pi(P_2) = [1, 2] \cup [\frac{11}{2}, 6]$; the point $x = \frac{3}{2} \in \pi(P_2) \setminus \pi(S_2)$ lies in the component $[0, 4)$ of $G \setminus \pi(S_2)$, whose frontier point 4 is in $\pi(\alpha_2)$. That the copies of $[1, 2]$ in F_1 and F_2 form a component $D \subseteq F_1 \cup F_2$ of the fold with $\widetilde{D} = [0, 4)$ is a hand argument; the fold is not constructed in the certificate. The conclusion of the step holds, and the sentence fails only for components that the restriction on p. 209 discards. Verified: `ho16_prop19`, `ho16_prop19_full`, with a case split.

Item 21. Moore 1916, Theorem 10

Proof defective; statement true. Counted. Ledger A.12.

"$R_1 R_2 \cdots R_k R$ is a simple chain" need not hold. Repaired by taking a shortest sub-chain (`fa_theorem10`).

Item 22. Moore 1916, Theorem 15, part III

Proof defective; statement true. Counted. Ledger A.12.

"since y precedes every link of C_n that contains P it follows that P_B precedes P" fails at stage n, because P_B can lie in the next link, which may contain P. Repaired by choosing n large; the two "it is clear" steps of part III are supplied (`fa_theorem15`).

Item 23. Moore 1916, Theorem 28, last step

Proof defective; statement true. Counted. Ledger A.13.

The last step: "the point-set $A_2C + CP + PD + DB_2$ contains as a subset an arc AXB satisfying

conditions (1), (2), and (3)". The arc CA_2 runs in R_A, which straddles J, so its part before J need not lie in M. Repaired by taking A as the first point of CA_2 outside M, which lies on J; the point P is not needed (`fa_theorem28`).

Item 24. Moore 1916, Theorems 33–34, Kline's proof, p. 150

Proof defective; statements true. Counted once. Ledger A.13.

Two proofs are printed; Kline's general proof has a gap: the first point of DC on CAB may be B. Filled (`fa_theorem33`, `fa_theorem34`). Theorem 34 is verified for any region.

Item 25. Moore 1916, Theorem 36

Proof defective; statement true. Counted. Ledger A.14.

The arc BC can leave R; it is cut at its first point outside R (`fa_theorem36`). Moore 1925 Theorem 19 needs, in addition, that the interior lies in R (`fa_theorem36_interior_subset`).

Item 26. Moore 1925, Theorem 11, pp. 421–422

Proof defective; statement true under the corrected definition (Item 30). Counted. Ledger A.16.

"If B is not identical with J then it is a proper subset of J. It easily follows, with the help of Theorems 10 and 6, that B is a simple continuous arc". Theorem 6 needs B connected, which is not shown. Repair: B lies in a sub-arc A of J missing an element of $J - B$, and A would separate G, against Theorem 9 (`frontier_eq_of_scc`, `m25_theorem11`).

Item 27. Moore 1922, Theorem 4, necessity, last step, p. 237

Proof defective; statement true. Counted. Ledger A.20.

The points Y_m, Y_n lie in R, not in M, so "there exists no connected subset of M of diameter less than ε that contains both Y_m and Y_n" proves nothing. Repaired within the proof: take points of $\bar{R}_n$ on the middle square and use $\varepsilon/2$ (`moore22_theorem4`).

Item 28. Whyburn, *Analytic Topology*, (9.1)

Proof defective; statement true. Counted. Ledger A.19.

The proof applies [I, (10.2)], which requires $N \cdot G$ compact, to $U \cdot (Q \cdot R_a + p)$, which need not be compact. The case $N \cdot G = N$ is not treated, and "$Q \cdot R_a + p$ is a generalized continuum" is asserted. The repair runs boundary bumping in a compact neighbourhood; the openness of R is not needed (`whyburn_9_1`).

Local defects not counted

These are defective constructions or reasons inside proofs whose step is nonetheless right, or is repaired in one line.

- Bing–Jones 1959, Theorem 3 (p. 175): even with $F(w, m)$ read for the printed $F(v, m)$, the chain D is not a chain as written: $A \cdot e_{j-1}$ can meet f_{v+1} and $B \cdot e_{r+1}$ can meet f_{w-1}, which are not adjacent in D. Repaired by merging (`bj_theorem3`).
- Bing–Jones 1959, Theorem 8 (p. 178): the printed e_k need not form a chain, since $e(a_i)_k$

is controlled only over I_{a_i} (argued, not certified). Repaired by restricting each piece to the fibres over I_{a_i} (`bj_theorem8`).

- Bing–Jones 1959, Theorem 9 (pp. 179–180): the condition on ε concerns subsets of Q, while the links of F are not subsets of Q; it secures neither the disjointness of links in different middles nor that every link falls under a rule. Repaired with links whose closures lie in links of E and a smaller mesh (`bj_theorem9`).

- Hoehn–Oversteegen 2016, Proposition 10, second part (p. 190): "If $C' \neq C$, then there must be a point $x \in \partial C' \cap C$. But since A' has consistent complement relative to B, we must have $x \in B$". Consistency gives $\partial C' \subseteq B$ only if $\partial C'$ meets B, which is not shown. Repaired by shrinking V; the formal proof avoids the step (`ho16_prop10`).

- Hoehn–Oversteegen 2016, Theorem 15, Claim 15.1, property (4) (p. 195): "because x is not an endpoint of G by (5), and $\pi(M' \cap V)$ is open for some neighborhood V of (x, y) by (3)". Here $(x, y) \in T$, and (3) speaks only of points of $M' \setminus T$; the openness comes from irreducible separation. The step is true. Also p. 198, (S4): an incomplete case split (a component around a cycle of G is in neither case; the same chain argument covers it); and p. 197: nothing excludes $b \in J_j$. The formal proof of Theorem 15 avoids all of pp. 194–198 (`ho16_theorem15`).

- Hagopian 1976, Lemma 4, first paragraph (p. 341): "Since the decompositions of E and F are minimal, each element of the decomposition of E (except possibly $k^{-1}(1)$) is an element of the decomposition of F." Minimality gives it only for layers in $d^{-1}(u)$ with $u < s$ (`hagopian76_lemma4_layers'`); a plane model without homogeneity has a layer swallowed. Verified in a strong form through Lemma 5 (`hagopian76_far_layer`).

- Hagopian 1976, ε_1 (pp. 345–346): "the chain formed by taking ε_1-neighborhoods … follows $h[E]$" does not follow from the four printed families; true after shrinking ε_1. Likewise property (D) of T9.

- Hagopian 1984, Lemma 15 (p. 341): "By Lemma 3" is insufficient as printed (the layer through a point of $h[Z]$ need not lie in $\Delta(\mathcal{P})$); verified with a smaller ε (`hagopian84_lemma15`). The cover (p. 343): "it follows that $\bigcup \mathfrak{B}$ is a closed open subset of M" is not an argument over uncountably many Z; a finite subcover is proved directly (`hagopian84_cover`, `sUnion_cover84_eq_univ`). Disjointness, first case: "by (2), (3), and Lemma 7, $A = B$" leaves the case $S_A \subseteq Y$ open; supplied by a transfer lemma.

- Oversteegen–Tymchatyn 1982, Corollary 3.5 (pp. 139–140): "there is a continuum $B \subset X$ such that $[0, 5/8] \subset \pi_1(h_K(B)) \subset [0, 3/4)$": the first inclusion is impossible as printed, since $h_K(B) \subset K \subset [\varepsilon, 3/4] \times R$. Of the two readings only $[0, 5/8] \subset \pi_1(B)$ gives what Theorem 3.2 needs; the existence of B is unargued and is supplied through hereditary indecomposability. "Without loss of generality … $\pi_1(A) = [0, 1/2]$ and $\pi_1(X) = [0, 1]$" is not a rescaling; supplied by a shear and a homeomorphism.

- Jones 1975, (d), second sentence (p. 58): "c' has the properties possessed by c" holds only if the Effros homeomorphism preserves the direction of T; adding $d(a, ce)$ to the minimum repairs it. The triod paragraph does not yield the contradiction as printed.

- Fugate 1969, Lemma 3: Lemma 2 gives $K \cup R$ irreducible between some $q \in K$ and some p; getting $p \in R \setminus K$ needs $R \not\subseteq K$, which is not argued. Supplied from Sorgenfrey's theorem.

8 Statements false because of their wording

In each item a definition or hypothesis says less than the author meant, and a configuration the author plainly did not have in mind refutes the printed statement. The intended statement is verified in each case.

Item 29. Moore 1925, the definition of complementary domain (p. 417) and Theorem 2

False as printed. Counted (one false statement). Ledger A.16.

"A domain of elements of G is said to be a complementary domain of a closed set H … provided the boundary of D is a subset of H." Neither disjointness from H nor maximality is required, so G is a complementary domain of every closed set, and "the unbounded complementary domain" is not unique. Theorem 2 is false under this definition (`not_m25_theorem2_printed`): take the decomposition into points, M a closed disk, D the open disk minus a radius, and E the plane minus the radius; the outer boundary is the radius, which is cut by each of its interior points. Read as components of the complement, Theorem 2 holds (`m25_theorem2`), and the unbounded one is unique (`m25_unbounded_unique`).

Item 30. Moore 1925, the definition of simple closed curve (p. 417) and Theorems 10 and 11

False as printed. Counted (two false statements). Ledger A.16.

"A bounded subset H of the set G is said to be a simple closed curve … provided it is disconnected by the omission of any two of its elements." Neither closedness nor connectedness is required, nor more than one element. Two disjoint segments (decomposition into points) qualify; for them Theorems 10 and 11 fail: they are not a union of two arcs, and their complement is connected (`not_m25_theorem10_printed`, `not_m25_theorem11_printed`). With "closed, connected, nontrivial" added, both are verified (`m25_theorem10`, `m25_theorem11`).

Item 31. Moore 1925, Theorem 7, p. 420

False as printed, by a misprint. Counted (one false statement). Ledger A.16.

"then p is a limit element of the set H" should read x. The printed form is refuted (`not_m25_theorem7_printed`); the corrected form is verified (`m25_theorem7`).

Item 32. Mullikin 1922, Theorem 5, p. 160

False as printed. Counted (one false statement). Ledger A.17.

"… such that M_1 and M_2 have in common only K_1 and K_2, where K_1 and K_2 are mutually exclusive connected sets, then $S - M_1 - M_2$ is the sum of just two mutually exclusive, connected domains." Mutually exclusive connected sets need not be separated: take $M_1 = M_2$ a segment, K_1 a half-open half and K_2 the closed other half; their union is connected, and the complement of the segment is connected (`not_mullikin_theorem5_printed`). The intended hypothesis is that K_1, K_2 be closed; Moore's use ($K_1 = \bar{p}$, $K_2 = \bar{q}$) satisfies it. Verified in that form (`mullikin_theorem5`), by logarithms and Borsuk's theorem instead of polygons.

Omitted hypotheses and definitions, not counted

- Hoehn–Oversteegen 2016, pp. 193–194: "if k is odd then $\pi(S) = G$" and "π maps S homeomorphically onto G" omit "G connected"; with the paper's convention ("G will denote a (not necessarily connected) graph", p. 184) both fail for $G = [0,1] \cup [2,3]$, $S = [0,1] \times \{\frac{1}{2}\}$ (`ho16_p193_remark_false`).
- Hoehn–Oversteegen 2016, Proposition 17 (p. 200): the hypothesis $k \geq 3$ is not printed (for $k \leq 2$ the conclusion has height ≤ 0); harmless, since Theorem 20 applies it to odd $k \neq 1$.
- Moore 1916, Theorem 31: the order $C\,E\,B\,G\,D$ needs $B \neq C, D$. Theorem 32: nothing keeps $C \in R$ off the arc AB; proved under $R \cap AB = \emptyset$. "The interior of J" is used from Theorem 23 on and never defined; supplied as the region bounded by J.
- Moore 1919: the double-ruling definition (p. 172, footnote) requires equal numbers of arcs in the two families, which Moore's own Fig. 1 (9 and 8) violates; padding shows the requirement costs nothing. It omits the disjointness of the Y-arcs; the "subdivisions" are never defined.
- Moore 1925: the definition of boundary (p. 417) is empty in its second clause as printed; read "of G", it is the frontier. Theorem 3's statement does not say that the region contains p, which the proof gives and Theorem 4 needs.
- Moore 1920, Definition 4 (p. 342) admits a single point; see Item 35.
- Hagopian 1976, p. 340: the end-set property "[14, Theorem 8, p. 14]" is true for the decomposition and false for an arbitrary admissible map.
- Mullikin 1922, Theorem 4 (p. 154): true for disjoint M_1, M_2 under her convention that the empty set is connected (`mullikin_theorem4_disjoint`), false under Mathlib's (`not_mullikin_theorem4_disjoint_mathlib`).

9 Statements false at degenerate cases

Each statement in this section fails only at an empty set, a single point, an endpoint, or a chain of at most three links, and holds once that case is excluded. In most of them the proof begins by choosing a point that the hypotheses do not supply. Bing 1948 Theorem 4 is different: its proof claims the small case as easy (Item 38). Moore 1916 Theorem 43 has no printed proof.

Item 33. Jones 1948, Theorems 6 and 10

False as printed. Counted (two false statements). Ledger A.10.

Theorem 10 (p. 407): "*In order that a compact continuum M be indecomposable it is necessary and sufficient that M contain an open subset D of M such that if x is a point of D and y is a point of $M - x$, then M is not aposyndetic at x with respect to y.*" $D = \emptyset$ satisfies the condition vacuously in every continuum, so the statement says every continuum is indecomposable; an arc refutes it (`theorem10_printed_iff`, `not_theorem10_printed_unitInterval`). Theorem 6 has the same defect: with $D = M_0 = \emptyset$ the hypotheses hold vacuously and the conclusion asks for a point of D (`not_theorem6_printed`). Both are verified with D nonempty. Theorems 11, 12, 13 and 19 carry the same tacit hypothesis and are not listed as false, because the intended reading is plain.

Item 34. Moore 1916, Theorems 4, 14, 19, 23, 35 and 43, and the limit-point criterion of p. 139

False as printed. Counted (seven false statements). Ledger A.12–A.14.

- Theorems 4, 19, 23 and 35: adjoining $\emptyset$ to the regions preserves Axioms 1–8 and every derived notion, and each statement fails for the empty region. Moore's proofs begin by choosing a point of R. Verified for R nonempty (`fa_theorem4`, `fa_theorem19`, `fa_theorem23`, `fa_theorem35`).
- Theorem 14: the proof opens "For each n, M_n contains at least one point", which is not a hypothesis; $M_n = \emptyset$ refutes it (`not_fa_theorem14_printed`); verified with M_n nonempty (`fa_theorem14`).
- Theorem 43: false for $G = \emptyset$ in every model (`not_fa_theorem43_printed`). It has no printed proof; the corrected form is verified through a dual of Theorem 41 for intersections of interiors (`fa_theorem43`).
- The limit-point criterion on an arc (p. 139, unnumbered) fails at an endpoint, where it holds vacuously (`not_fa_sec5_limitPt_printed`). Corrected and verified.

Item 35. Moore 1920, Theorem 4

False as printed. Counted (one false statement). Ledger A.16.

Definition 4 (p. 342) of a simple closed curve has "closed, connected, and bounded" but admits a single point, for which Theorem 4 is false (`not_moore1920_theorem4_singleton`); so is Moore 1925 Theorem 11 (`not_m25_theorem11_singleton`).

Item 36. Jones 1949, Theorems 1 and 2

False as printed. Counted (two false statements). Ledger A.18.

Theorem 1: "If a compact, plane continuum M is both homogeneous and aposyndetic, then M is a simple closed curve" fails for a one-point continuum (`not_jones49_theorem1_point`). Jones's footnote 3 refers the reader for definitions to Moore's book or Whyburn's; Whyburn's convention (*Analytic Topology*, p. 15: "A compact connected set will be called a continuum") admits one-point continua. Theorem 2 fails likewise: a point is homogeneous and has no weak cut point (`not_jones49_theorem2_point`). Both are verified for nondegenerate continua (`jones49_theorem1`, `jones49_theorem2`).

Item 37. Hoehn–Oversteegen 2016, Theorem 1′, p. 212

False as printed. Counted (one false statement). Ledger A.26.

Theorem 1′ ("surjective semispan zero" in place of "span zero") omits "non-degenerate" and fails for a one-point continuum (`ho16_theorem1prime_printed_false`). It is stated correctly nowhere, although p. 212 presents it as what "our proof of Theorem 1 in fact establishes". Theorem 1 as restated on p. 211 omits "non-degenerate" too (`ho16_theorem1_printed_false`); it is recorded as a misprint, because p. 180 states it correctly.

Item 38. Bing 1948, Theorem 4, at $m = 3$, p. 733

False as printed for $m = 3$; true for $m \geq 4$. Counted (one false statement). Ledger A.1.2.

"THEOREM 4. *If $D = [d_1, d_2, \cdots, d_n]$ is a chain crooked in the chain $E = [e_1, e_2, \cdots, e_m]$ and $D(r, s)$ is a subchain of D such that a link of $D(r, s)$ intersects e_1 and a link of $D(r, s)$ intersects e_m, then there is a chain F such that F is a consolidation of D, each element of F is a subset of the sum of two adjacent links of E, d_r is contained in only the first link of F, and d_s is contained in only the last.*" The proof: "Since the case of our theorem where $m \leq 4$ is easily established". Two failures at $m = 3$. With $r = s$: three intervals in $\mathbb{R}$ crooked in three intervals, the middle fine link meeting the first and last coarse links; d_r in both the first and the last link of F forces F to have one link, all of D, which lies in no two adjacent links of E (`theorem4_two_degenerate_false`, `not_theorem4Literal`). With $r < s$: in $\mathbb{N}$ with the discrete topology, coarse links $\{0, 1, 5\}$, $\{1, 2, 4, 5\}$, $\{2, 3, 4\}$, fine links $\{5\}$, $\{4, 5\}$, $\{2, 3, 4\}$, $\{1, 2\}$, $\{0, 1\}$, $r = 0$, $s = 1$ (`not_theorem4LiteralND`). Verified for $m \geq 4$ in every clause (`theorem4_literal_ge4`).

Item 39. Bing–Jones 1959, Theorem 7, second sentence, p. 177

False as printed for chains of one or two links. Counted (one false statement). Ledger A.25 (third reading).

"*In fact, if A, B are closed sets in $P \cdot d_1$, $P \cdot d_n$ respectively and $H(x_r) = d_1$, $H(x_s) = d_n$, the chain E may be selected so that $A \subset e_r$, $B \subset e_s$.*" With $n = 1$, $m = 3$, H constant, $A = B = \{p\}$, $r = 1$, $s = 3$: $p \in e_1 \cap e_3$, and e_1, e_3 are not adjacent (`bj_theorem7_ends_printed_false`). With $n = 2$ the same, with $p \in P \cdot d_1 \cdot d_2$ and $H = (d_1, d_1, d_2)$ (a hand argument). For $n \geq 3$ it holds when "closed" means closed in the space, which is Bing and Jones's own use on p. 178 (`bj_theorem7_ends`). Read as closed in the subspace $P \cdot d_1$, it fails for every n (a hand argument: $H = (d_1, d_2, d_1, d_2, d_3)$, $r = 1$, $A = P \cdot d_1$).

Tacit nonemptiness, not counted

Jones 1948 Theorems 11–13 and 19 (D or U nonempty); van Mill 2004 Proposition 2.2 ($X \neq \emptyset$; in the empty space $\emptyset$ is analytic and fat); Moore 1916's definition of connected (the two parts must be nonempty); Moore 1925 Theorem 12 (a domain read as nonempty); Cook 1967 Theorem 4 (the sequence C_n does not exist when $C = X$, a case that needs no argument); Whyburn (8.3) ("no convergent sequence of components" must exclude the constant sequence).

10 Unargued steps and statements without proof

Every step listed here is true and is supplied, or shown unnecessary, in the formalization, except those marked "not decided". Sources are listed in the order of §2.

Item 40. Jones 1948, 1949, 1951 and 1955

Unargued; stated without proof. Ledger A.10, A.11, A.18.

- Jones 1955, Theorem 1: the continuity step ("Since G must be continuous at some one of its elements": an upper semicontinuous decomposition of a compact metric space is continuous at a residual set of elements); homogeneity and aposyndesis of the decomposition space in (a); "N_o is closed". Not decided and moot: the attainment of the supremum $d(q)$ and "E_n is closed".
- Jones 1955, Theorem 2: "if one element of G were to separate S, then each of them would"

uses Borsuk's theorem; the "uncountably many mutually exclusive domains" (their choice and disjointness); the upper semicontinuity of G' and "G is a compact subcontinuum of G'"; "$M - g$ is connected"; the aposyndetic case ($L_x = \{x\}$) is not mentioned. Verified (`jones55_theorem2'`). The three unnumbered claims of the closing remarks are verified.

- Jones 1951, Theorem 1: "M' is homogeneous and aposyndetic", "M' contains a nonseparating point" and "M' is hereditarily unicoherent" are asserted; the formal proof needs none of them.
- Jones 1949: Case 1 (that every point of M belongs to the boundary of an element of G needs the Jordan curve theorem, `jordan_curve_theorem`); four steps of Case 2, the last of which ("$M - (A + B)$ is *not* connected") needs θ-curves and Janiszewski's theorem; and "all except countably many … must be of order 2" needs M uncountable.
- Jones 1948: Theorems 0, 2, 3 and 5 are stated without proof (all proved); Theorem 13's choice of V_2 must satisfy $\overline{V_2} \subseteq W_1$.

Item 41. Whyburn 1939 §14, Whyburn 1942, Moore 1922, Menger 1932

Unargued; stated without proof. Ledger A.19–A.21.

- Whyburn 1939, Theorem 14: "N can intersect only one of the arcs xuy, xvy"; "the arc x_1xuyy_1 of J is contained in C"; the limit continuum H lies in M and is nondegenerate; uniform continuity in the finite-cover step. Not decided, not needed: "Hence there exists a simple closed curve K in $V_\varepsilon(p)$ such that $K \cdot M = c + d$". Corollary 1 and (6.22) are stated without proof (supplied); footnote 16 (equivalence with Wilder's local 0-avoidability) is not decided and not used.
- Whyburn 1942: (8.4) "there exist points x and y of N … such that $\delta[Q_2(x) \cdot Q_1(y)] < \varepsilon/4$" needs facts from inside the proof of (2.2), not from the statement of (3.2); (8.41) uses the boundary bumping of (9.1); (9.2) "and hence also of order 2 in L". (3.1) is stated without proof (not used).
- Moore 1922, Theorem 4: five steps of necessity and three of sufficiency, among them the sector theorem for continua crossing an annulus (proved by lifting to a strip) and the cyclic order of the continua k_n.
- Menger 1932, p. 266: three steps of the *Kennzeichnung der gewöhnlichen Kurven*; not used, since the formal proof follows Menger's second argument on p. 267.

Item 42. Hagopian 1975, 1976 and 1984

Unargued. Ledger A.22–A.24.

- Hagopian 1975: Lemma 1, three steps (the triod must be proper; complementary domains missing M are pairwise disjoint; Baire and Moore 138); Theorem 1, paragraph 2 ("F, $f[F]$, and $g[F]$ are mutually disjoint"; "$B \cup F \cup f[F] \cup g[F]$ is a triod"); paragraph 9, the second inclusion of "$h_i[Y] \subset h_i k^{-1}[[r, t]] \subset E$".
- Hagopian 1976: Lemma 4, the second step (about two hundred lines of the formal proof, `exists_proper_extension`); Lemmas 6, 7 and 9 (several sentences each); the plane Schoenflies theorem in Lemma 9 (`plane_schoenflies`); "we can assume … that $k^{-1}[(0, 1)]$ is accessible"; Properties 1 and 3; T7, T9, T11, T12, T13; "It follows from Property 3 that I is finite"; "Note that $x_1 = 0$, $x_2 = e$, and $x_{2n} = 1$"; Property 4 for the top block; the order of the constants,

which the text does not fix. Lemma 5's union of two extensions is not decided and not used. Every one of these is supplied except those listed in Item 3.

- Hagopian 1984: Lemma 7 (three atriodicity steps, avoided); Lemma 8 (an arc); Lemma 12; Lemma 13 (the hypotheses of Lemma 3, and an orientation argument for regularity); Lemmas 14–15 (positivity of $\varepsilon = \frac{1}{2}\rho(\bigcup \Delta(\mathcal{P}), M \setminus \bigcup \mathcal{P}')$); Lemma 16 ("Since $\mathcal{P}$ is free, there exist a 3-linked subchain …", replaced by a different lemma; Lemma 16 without normality is not decided); in Theorem 1, $X(Z)$, $J(Z)$ (without the printed positions of its end sets), claims (5)–(6), and "$M/\mathfrak{D}$ … contains an arc".

Item 43. Jones 1975

Unargued. Ledger A.28.

(c) (the argument from (a) and (b) is not given). In the Theorem: "let us assume that T is accessible from the complement of M"; "There exists a positive integer η … This follows from property (f)"; "Clearly t_f cannot lie in $\bar{D}_1 \cup \bar{D}_2 \cup \bar{D}_3$ without violating the selection of η"; "it is easy to see that $p \notin \bar{D}_2$ and property (f) is violated". None is used by the formal proof (Item 4).

Item 44. Oversteegen–Tymchatyn 1982 and Cook 1967

Unargued. Ledger A.29.

- Theorem 3.6: "Since X is atriodic tree-like" (not needed in the indecomposable case, so neither Rogers 1981 nor Jones 1969 is needed for Theorem C); "Hence there exists an index i and points … such that $d(x, y) < \delta$"; "since Z is a proper subcontinuum".
- Corollary 3.5: "For each $K \in \mathcal{K}$ we may suppose $K \cap \{\varepsilon\} \times R = \{x_K\}$" (minimality does not give it in general; not needed); the uncountable family $\mathcal{K}$ of disjoint crossing continua.
- Theorem 2.7: "there exists a continuum H in C such that $(0, p) \in H$ and H meets both $I \times \{0\}$ and $I \times \{1\}$" and "Clearly $K \cap H \neq \emptyset$"; the continuity of φ can fail where the path meets B_1 (harmless where it is used). Remark 2.5 and the sketch of Lemma 2.1.
- Cook 1967, Theorem 4: the sequence C_n is not constructed, and "The common part of $f(C_1)$, $f(C_2)$, … is $f(C)$" is unstated; the formal proof needs neither (`cook67_theorem4`).

Item 45. Hoehn–Oversteegen 2016

Unargued; stated without proof. Ledger A.26.

- p. 211, proof of Theorem 1: "It follows that $d_H(X, T) < \delta$"; from $d_{\sup}(f, \mathrm{id}_X) < \delta$ only $X \subseteq T_\delta$ and $f(X) \subseteq X_\delta$ follow, while Theorem 4 needs $T \subseteq X_\delta$ too. Repair: replace T by $f(X)$.
- Theorem 8, (1)⇒(2): the fans must be pairwise disjoint, which is not said.
- Lemma 6 (p. 185): "The proofs of these properties are left to the reader"; seven properties, verified (`ho16_lemma6`).
- pp. 186–187: the fold determined by G_1, G_2, G_3 ("It is straightforward to see") and Proposition 7 (three steps, supplied).
- pp. 189–191: the observations after Definition 9; Proposition 11 (three short steps).

- pp. 194–198, Theorem 15: Claim 15.1 ("We leave it to the reader"); the passage to an irreducible subset; the branch-point surgery; "clearly S irreducibly separates"; "(S3) and (S5) are immediate". The formal proof avoids all of them (`ho16_theorem15`).
- Lemma 18 (that $\pi(S')$ has finitely many components); Proposition 17 ("It is now easy to verify"); pp. 203–210 (the "straightforward observations", (S3) and (S5′) in Proposition 19, the component K of p. 209).

Item 46. Bing–Jones 1959

Unargued. Ledger A.25.

- p. 173: "It follows from Theorem 10 that any two circles of pseudo-arcs are topologically equivalent and that each is homogeneous"; p. 189: "It follows from Theorem 10 that each circle of pseudo-arcs is homogeneous and that any two of them are homeomorphic." The passage from arcs to circles is not written; it needs circle-likeness to make the two halves snake-like (`isSnakeLike_of_subset_circleLike`), and then Theorem 10 and a gluing.
- Theorem 2 ($G(b,c)$ meets d_t); Theorem 3 (D an ε-chain); Theorem 7 (above); Theorem 8 (U_A, U_B, asserted for an upper semicontinuous collection; the cover $U_1, \dots, U_t$, "It follows from Theorem 4"; the general case, "alter Y as was done in the proof of Theorem 7"); the corollary to Theorem 8 (three consecutive links); Theorem 9 ("That h is a homeomorphism of P onto Q follows from an argument similar to").
- Theorem 10: Step 1(b); Step 2(b), the chain map $R_{i/n}$; Step 2(c), Theorem 6 applied without naming the chain in the role of E, the use of the false Theorem 5 (Item 8), and the transition set the repaired Theorem 5 needs; "Of course, $d(p, i+1) \subset d(p, i)$" (p. 188, from Condition B); Step 2(i), Theorem 9's hypothesis; Step 3 (closures of the f_j); "Steps 4, 5, ···. These steps are essentially repetitions of Step 3." Verified (`bjThm10`, through `bj_step`).

Item 47. Bing 1948 and 1951; Fugate 1966 and 1969; Lehner 1961; van Mill 2004

Unargued; stated without proof; not established for the generality stated. Ledger A.1–A.9.

- Bing 1948, Theorem 14: "That there exist such sequences follows as a consequence of Theorem 9" is not a proof; the descriptions are supplied by Lehner's Lemma 3 for $n = 2$ (`lehner_lemma3_two`). Two assertions of the sketch are not established for an *arbitrary* covering description; both hold for the description Lehner's construction produces. Nine further steps are supplied.
- Bing 1951 (Duke): Theorem 7 is stated for tree-like continua, but the proof of Theorem 6 that it rests on assumes that no subcontinuum separates the plane; "tree-like ⇒ no separating subcontinuum" is not stated (verified, `treeLike_nonseparating`). Theorem 8: "K_2 contains K_4 interiorly" and the order of the two inclusions (`theorem8_1951'`). Theorem 11, sufficiency: three sentences, supplied from Fugate 1969 (`theorem11_1951`).
- Fugate 1966, Corollary 1: stated without proof; the induction it suggests needs every partial union to be unicoherent, which is what is missing; verified (`fugate_cor1`, `fugateU_holds`). No proof or discussion of the corollary for more than two pieces was found in the reviews, in later work of Fugate, in Nadler 1992, Macías 2005, the Houston Problem Book, or the citing papers searched.

- Fugate 1969: besides Lemma 3 (§7), nine smaller steps (hereditary unicoherence not stated; the displayed chain is only a weak chain).
- van Mill 2004, Proposition 3.5: Proposition 2.2 is applied "in E", which needs "nowhere meager in X" to give "nowhere meager in E"; avoided by a relative form of 2.2.

Item 48. Moore 1916, 1919 and 1925

Unargued; stated without proof. Ledger A.12–A.16.

- Moore 1916: Theorems 1, 3, 7, 11, 13, 16–18, 20, 22, 30, 38, 40, 42 and 43 are stated without proof (all proved; Theorem 1 needs no axiom); §5 rests on Lennes 1911 and supplies no proofs; Theorem 41's last step ("It may be proved that J is a closed curve satisfying the requirements") hides a limit construction; Theorems 37 and 39 (about seven steps). The opening of §6 ("It is clear that Theorems 1–19 are consequences of Axioms 2–5 and Theorems 5 and 15") is partly borne out and not formalized.
- Moore 1919: Theorem B (five steps called "clear" or "easy"); Theorems D and E ("This process may be continued", and the extension of rulings, never proved); Theorem E's limit arcs; the nine-cell ruling of Fig. 3 (not needed); Theorem G ("It is clear that through every point there is just one curve of the set α …"); the final paragraph ("It is clear that the correspondence thus established … is continuous"). Verified by a route through a Schoenflies theorem for Moore's space (`m19_theoremG`, `m19_plane`). The claim of the introduction for Σ_2 rests on Moore 1916 p. 163 (Item 12).
- Moore 1925: Theorem 2 (Brouwer's theorem is needed twice, and three "clearly" steps); Theorem 11 ("therefore neither $\bar{a}$ nor $\bar{b}$ separates S"; "It is easy to see that one of these domains … is bounded"; "Clearly I and E are domains … and $I + E = G - J$"); Theorem 3 (four steps); Theorems 4, 5, 6, 9, 10 and 15–19 and the p. 420 claim, stated without proof (supplied); Theorems 8, 13, 14 and 20 (two to three steps each).

11 Misprints and harmless mismatches

Each is read as indicated; none affects a result. "Printed X, read Y."

- **Hoehn–Oversteegen 2016.** p. 211, Theorem 1 restated without "non-degenerate" (Item 37); p. 189, Theorem 8, (3)⇒(1): U and V interchanged in the first sentence; p. 195: x for (x, y); p. 198: "Since $a_i \in C$" for a_p, and $\{z\}_*$ for the star; p. 202: $\cap$ for $\cup$ in "$\mathcal{E}(S'') \subset (F_1 \cap F_2)_\star$", and "$S''$ is a straight" (a word missing); pp. 205–207: φ^{-1} for $\varphi_\star^{-1}$, three times, and on p. 205 "$P_1' \cap (\partial F_3)_\star = \varphi_\star^{-1}(\gamma_{i_0})$" for its intersection with $(\partial F_3)_\star$; p. 209: the order of composition; p. 192: "comple ment"; p. 210 says more than p. 201; Theorem 20 (p. 210): an unneeded assumption on the ε_i; pp. 202–203: two omitted citations of (S4′).
- **Bing–Jones 1959.** p. 172: *of* for *if*; p. 173: *of* for *are*; Theorem 3: $F(v, m)$ for $F(w, m)$; Theorem 7: "components" for "composants"; Theorem 8: d_2 for d_n; Theorem 9: G for F, and e for d on p. 180; p. 181: E_i for E_1, H_2 for H_1, d_k^1 for e_k^1; p. 182: K_3 for K_2; p. 185: $R_{(2i+1)/n}$ for $R_{(2i+1)/2n}$ and $P(0, 1/n - \varepsilon)$ for $P(0, i/n - \varepsilon)$; p. 186: E_k for E_1; p. 187: "intesects", E_1 for D_1, f_{i+1} for f_{t+1}; p. 178, the corollary: $4i+1, \dots, 4i+4$ for $4i-3, \dots, 4i$; p. 188: $H_i(d_k^{i+1})$; p. 189: $\rho(p, q) < 4/2^i$ where the meshes give $8/2^i$; p. 180: the last display of Theorem 9 ends "$\subset \bar{e}_i$" where the claim is $h(P \cdot d_i) \subset e_i$; p. 187: condition (1) asks for the whole link f_j

where the construction controls only $f_j \cdot Q(i/k,(i+1)/k)$; p. 191, the figure: the level written second, and a_1, c_1 for a_i, c_i; pp. 190–191: the letter G used twice.

- **Hagopian.** 1975: "transfromation" (p. 37); Lemma 3 printed for type A, citing a source not in hand (only type A′ is used). 1976: none beyond the citations of Item 17. 1984: "opens sets" (p. 333), "ε-circluar" (p. 334), "ε-homomorphisms" (p. 338), unbalanced brackets in Lemma 13(5), Lemma 13(5) printed without h (p. 340), $K(Z)$ "the component of $\bigcup \mathcal{P}'$ that contains Z" (read: whose closure has Z as a layer); Lemma 2 (cited to Mackowiak–Tymchatyn 13.8) is applied to unions $F \cup J$, $N \cup F$, which is harmless in the plane.
- **Oversteegen–Tymchatyn 1982.** "$\sigma_0(x)$" (p. 129); "$f(X) \supset g(X)$" for $f(C) \supset g(C)$ and "homomorphism" (p. 140); "[7]" for Effros, which is [6] (pp. 139–140); R^2 for I^2 (p. 136); "$d(h_\lambda(Z), h_\gamma(z)) < 1/8$" for $h_\lambda(z)$ (p. 138); "an indecomposable homogeneous plane" ("continuum" missing).
- **Jones.** 1948, Theorem 16: P for the weak cut point p. 1951: "hereditorily" (p. 858). 1955, Theorem 2's proof: g for G.
- **Moore.** 1916: p. 139, "between P_1 and B" must allow B; Theorem 33, an unbalanced parenthesis; Theorem 34, "Jordan *arc*" for "closed curve". 1919: footnote † (p. 170), A and B for X and Y. 1922: Theorem 1, K^* for $\bar{K}$; Theorem 3, "diameter less than ε" where only $\leq 2\varepsilon$ holds (harmless with $\varepsilon/3$); Theorem 4, t_m for $\bar{t}_m$, $1 \leq m \leq n$ for 4^n, and $\varepsilon = (b-c)/4$ for $(c-b)/4$; "fie", "curce", "a sumed". 1925: Theorem 1 (two slips); Theorem 14, pxq with x never introduced; Theorem 18, x and y not introduced.
- **Whyburn 1939.** "$ou - o$" for $ou - u$; "$< 3\varepsilon$" after "$< 4\varepsilon$" (the argument gives 2ε); "there exists points".
- **Fugate 1966.** The introduction says "a-triodic and hereditarily unicoherent", Theorem 1 "a-triodic and unicoherent" (equivalent for two pieces).
- **Domokos–Prajs 2022.** "homogenous" (p. 13).

A Readings refuted that are not defects of the sources

Each of the following is a reading of a printed statement that the formalization refuted, and that the source does not make. They are recorded so that the refutations are not mistaken for defects.

Item 49. Bing 1951 (Duke), Theorem 8, p. 658, with g_p read with one layer

A misreading, not a defect: Theorem 8 is true as printed. Not counted. Ledger A.2.1, B.1.

Bing defines (p. 658): "For each point p let g_p denote the intersection of all continua which contain interiorly a continuum that contains p interiorly." With the inner continuum dropped, g_p is the intersection of the continua containing p interiorly, and Theorem 8 fails: in the plane continuum S_0 made of two sine curves over a bar, $q \in g_p$ while $g_p \neq g_q$ (`Counter92.theorem8_1951_false`). With Bing's definition every clause is verified (`theorem8_1951'`), clause (a), "G is an arc with respect to its elements", in the form Bing's proof establishes: of any three elements, one separates the other two; on S_0 the elements are the bar and the points of the curves (`Counter92.bingElt_bar`, `Counter92.bingElt_curve`). The inner layer is what makes the elements saturate.

Item 50. Other refuted readings

Not defects. Ledger B.1.

- Bing 1948, Theorem 3, end clause read "respectively": false at $k = 3$ (`Recon121.not_theorem3Literal`); Bing's own "or" is proved.
- "1951-crooked implies 1948-crooked": the two papers' notions of crookedness differ (`isCrookedIn_ne_isCrookedIn'`; four strips and six rectangles).
- Bing 1951, Theorem 11, the separation read with $K' \supseteq K - g' - N$: refuted.
- Nadler, *Continuum Theory*, 11.28 read without $U - H \neq \emptyset$: false for $H = \bar{U}$ (`isConnected_diff_of_superset_false`); Nadler's proof assumes it.
- van Mill 2004, Lemma 3.4 and Proposition 3.5 read without their closure bars: the printed statements, with the bars, are right.

B Lean names

Every name cited in the text, with its full name and the axioms its audit lists in `axioms.txt`. "Std" is `propext`, `Classical.choice`, `Quot.sound`; a dash means no axioms. "Used" marks the declarations that the two final theorems use (the dependency census, `doc/depcensus/used.txt`). The last column gives the items that cite the name.

Name	Axioms	Used	Items
`HPC.BJ59.bj_step`	Std	yes	46
`HPC.BJ59.bj_theorem3`	Std		§7
`HPC.BJ59.bj_theorem5`	Std		8
`HPC.BJ59.bj_theorem5_printed_false`	Std		8
`HPC.BJ59.bj_theorem5_printed_false_general`	Std		8
`HPC.BJ59.bj_theorem7`	Std		19
`HPC.BJ59.bj_theorem7_ends`	Std		39
`HPC.BJ59.bj_theorem7_ends_printed_false`	Std		39
`HPC.BJ59.bj_theorem7_proof_n3`	Std		19
`HPC.BJ59.bj_theorem8`	Std	yes	§7
`HPC.BJ59.bj_theorem9`	Std		§7
`HPC.BJ59.bjPlaneExample`	Std	yes	2
`HPC.BJ59.bjThm10`	Std	yes	1, 46
`HPC.BJ59.circ_step`	Std	yes	2
`HPC.BJ59.CircDescr.isCircleLike_M`	Std	yes	2
`HPC.BJ59.CircDescr.isCircleOfPA`	Std	yes	2
`HPC.BJ59.dp22_theorem61'`	Std		1
`HPC.BJ59.dp22_theorem61_iff'`	Std		1, 14
`HPC.BJ59.exists_circDescr_plane`	Std	yes	2
`HPC.BJ59.ho16_classification_plane`$_4$	Std	yes	§2, 1
`HPC.BJ59.ho16_classification`$_4$	Std	yes	§2, 1
`HPC.BJ59.hoCircleLike'`	Std		1
`HPC.BJ59.isSnakeLike_of_atomic'`	Std	yes	1
`HPC.BJ59.sweep_lemma`	Std	yes	2
`HPC.BJ59.sweep_lemma3`	Std		2
`HPC.BJ59.theorem5_repaired`	Std	yes	8
`HPC.Fold.not_theorem4Literal`	Std		38

Name	Axioms	Used	Items
HPC.Fold.not_theorem4LiteralND	Std		38
HPC.Fold.not_theorem8_literal	Std		9
HPC.Fold.Recon121.not_theorem3Literal	Std		50
HPC.Fold.Recon121.seam_every_H_fails	Std		6
HPC.Fold.Recon121.seamD_special	Std		6
HPC.Fold.Recon121.theorem3LiteralOr_holds	Std		6
HPC.Fold.Recon122.bing_main	Std		6
HPC.Fold.Recon122.special_five	Std		6
HPC.Fold.theorem3_bing	Std		6
HPC.Fold.theorem3_literal	Std		6
HPC.Fold.theorem4_literal_ge4	Std		38
HPC.Fold.theorem4_two_degenerate_false	Std		38
HPC.Fold.theorem7_minimal	Std		9
HPC.Fold.theorem8	Std		9
HPC.Fold.theorem9	Std		9
HPC.Hagopian.bing60_plane_uncond	Std		4
HPC.Hagopian.blockLayersMeetSides3	Std		3
HPC.Hagopian.blockLayersMeetSides_false	Std		3
HPC.Hagopian.exists_proper_extension	Std	yes	42
HPC.Hagopian.exists_separating_subcontinuum_of_homeomorph	Std	yes	17
HPC.Hagopian.hagopian76_far_layer	Std		§7
HPC.Hagopian.hagopian76_lemma4_layers'	Std		§7
HPC.Hagopian.hagopian76_theorem1	Std	yes	3
HPC.Hagopian.hagopian76_theorem1_of_cases4	Std		3
HPC.Hagopian.hagopian84_cover	Std	yes	§7
HPC.Hagopian.hagopian84_decomp	Std	yes	3
HPC.Hagopian.hagopian84_lemma15	Std	yes	§7
HPC.Hagopian.hagopian84_lemma7	Std		3
HPC.Hagopian.innerComponentsCross_hagopian	Std		3
HPC.Hagopian.jones75_contra	Std	yes	4
HPC.Hagopian.jones75_noArc	Std	yes	4
HPC.Hagopian.JonesSetting.jones75_d	Std		4
HPC.Hagopian.JonesSetting.nested	Std	yes	4
HPC.Hagopian.JonesSetting.orientation	Std	yes	4
HPC.Hagopian.JonesSetting.union_arc	Std	yes	4
HPC.Hagopian.last_step_ending	Std		3
HPC.Hagopian.no_proper_subcontinuum_separates	Std	yes	17
HPC.Hagopian.plane_schoenflies	Std		42
HPC.Hagopian.sUnion_cover84_eq_univ	Std		§7
HPC.Hagopian.trap_jordan	Std	yes	4
HPC.HO16.ho16_indec_homogeneous_iff	Std	yes	§2
HPC.HO16.ho16_lemma6	Std	yes	45
HPC.HO16.ho16_p193_remark_false	Std		§8
HPC.HO16.ho16_p208_false	Std		20
HPC.HO16.ho16_prop10	Std	yes	§7
HPC.HO16.ho16_prop19	Std	yes	20
HPC.HO16.ho16_prop19_full	Std	yes	20
HPC.HO16.ho16_theorem15	Std	yes	§7, 45
HPC.HO16.ho16_theorem1_plane	Std	yes	§2
HPC.HO16.ho16_theorem1_printed_false	Std		37
HPC.HO16.ho16_theorem1prime_printed_false	Std		37
HPC.HO16.ho16_theorem20	Std	yes	§2

Name	Axioms	Used	Items
HPC.isCrookedIn_ne_isCrookedIn'	Std		50
HPC.Jones.aposAt_of_isContinuum_interior	Std	yes	5
HPC.Jones.jones49_order_two	Std	yes	18
HPC.Jones.jones49_theorem1	Std	yes	36
HPC.Jones.jones49_theorem2	Std		36
HPC.Jones.jones55_a	Std	yes	5
HPC.Jones.jones55_b	Std	yes	5
HPC.Jones.jones55_c	Std	yes	1, 5, 14
HPC.Jones.jones55_continuous	Std	yes	5
HPC.Jones.jones55_elements	Std	yes	5
HPC.Jones.jones55_theorem2'	Std	yes	40
HPC.Jones.jones55_usc	Std	yes	5
HPC.Jones.jordan_curve_theorem	Std	yes	40
HPC.Jones.moore22_theorem4	Std		27
HPC.Jones.not_jones49_theorem1_point	Std		36
HPC.Jones.not_jones49_theorem2_point	Std		36
HPC.Jones.not_theorem10_printed_unitInterval	Std		33
HPC.Jones.not_theorem6_printed	Std		33
HPC.Jones.theorem10_printed_iff	Std		33
HPC.Jones.whyburn39_theorem14	Std	yes	11
HPC.Jones.whyburn_9_1	Std	yes	28
HPC.lehner_lemma3_two	Std		47
HPC.Moore.fa_theorem10	Std		21
HPC.Moore.fa_theorem14	Std		34
HPC.Moore.fa_theorem15	Std		22
HPC.Moore.fa_theorem19	Std	yes	34
HPC.Moore.fa_theorem23	Std		34
HPC.Moore.fa_theorem28	Std	yes	23
HPC.Moore.fa_theorem33	Std	yes	24
HPC.Moore.fa_theorem34	Std		24
HPC.Moore.fa_theorem35	Std	yes	34
HPC.Moore.fa_theorem36	Std	yes	25
HPC.Moore.fa_theorem36_interior_subset	Std	yes	25
HPC.Moore.fa_theorem4	Std	yes	34
HPC.Moore.fa_theorem43	Std		34
HPC.Moore.frontier_eq_of_scc	Std	yes	26
HPC.Moore.m19_plane	Std	yes	10, 48
HPC.Moore.m19_theoremC_corrected	Std		10
HPC.Moore.m19_theoremC_crosscuts	Std		10
HPC.Moore.m19_theoremC_within	Std		10
HPC.Moore.m19_theoremD	Std	yes	10
HPC.Moore.m19_theoremG	Std		48
HPC.Moore.m25_theorem10	Std	yes	30
HPC.Moore.m25_theorem11	Std	yes	26, 30
HPC.Moore.m25_theorem2	Std	yes	29
HPC.Moore.m25_theorem21	Std	yes	12
HPC.Moore.m25_theorem22	Std	yes	12
HPC.Moore.m25_theorem7	Std		31
HPC.Moore.m25_unbounded_unique	Std	yes	29
HPC.Moore.mullikin_theorem4_disjoint	Std		§8
HPC.Moore.mullikin_theorem5	Std	yes	32
HPC.Moore.not_fa_sec5_limitPt_printed	Std		34

Name	Axioms	Used	Items
HPC.Moore.not_fa_theorem14_printed	Std		34
HPC.Moore.not_fa_theorem43_printed	Std		34
HPC.Moore.not_m19_theoremC_printed	Std		10
HPC.Moore.not_m25_theorem10_printed	Std		30
HPC.Moore.not_m25_theorem11_printed	Std		30
HPC.Moore.not_m25_theorem11_singleton	Std		35
HPC.Moore.not_m25_theorem2_printed	Std		29
HPC.Moore.not_m25_theorem7_printed	Std		31
HPC.Moore.not_moore1920_theorem4_singleton	Std		35
HPC.Moore.not_mullikin_theorem4_disjoint_mathlib	Std		§8
HPC.Moore.not_mullikin_theorem5_printed	Std		32
HPC.Plane.Counter92.bingElt_bar	Std		49
HPC.Plane.Counter92.bingElt_curve	Std		49
HPC.Plane.Counter92.theorem8_1951_false	Std		49
HPC.Plane.fugate_cor1	Std		47
HPC.Plane.fugateU_holds	Std		47
HPC.Plane.isConnected_diff_of_superset_false	Std		50
HPC.Plane.isHereditarilyUnicoherent_union	Std		16
HPC.Plane.isSnakeLike_of_subset_circleLike	Std	yes	46
HPC.Plane.miller_1_5_false	Std		16
HPC.Plane.theorem11_1951	Std		47
HPC.Plane.theorem8_1951'	Std		47, 49
HPC.Plane.treeLike_nonseparating	Std		47
HPC.Span.cook67_theorem4	Std	yes	44
HPC.Span.ot82_cor35_of_thm32	Std	yes	7
HPC.Span.ot82_theorem32	Std	yes	7
HPC.Span.ot82_theorem36_of_cor35	Std	yes	7
HPC.Span.theoremC	Std	yes	7

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